

| Key                             | Function                                                |
|---------------------------------|---------------------------------------------------------|
| w                               | Toggle between shaded view and wire-frame view of model |
| Tab                             | Quick smooth shade preview                              |
| u                               | Auto-rotate view                                        |
| a                               | Focus view on selected elements                         |
| x                               | Align view to +x axis                                   |
| Shift+x                         | Align view to -x axis                                   |
| y                               | Align view to +y axis                                   |
| Shift+y                         | Align view to -y axis                                   |
| z                               | Align view to +z axis                                   |
| Shift+z                         | Align view to -z axis                                   |
| o                               | Toggle between perspective and orthogonal views         |
| Up, down, left and right arrows | Shift view up, down, left and right.                    |
| r                               | reset view to default view                              |

Table 2: View keys

keyboard short-cuts you should use, to get the maximum out of Wings 3D.

## Mouse functions

The left mouse button (LMB) selects elements, and turns them red. Clicking a selected element again deselects it. The middle mouse button (MMB) activates camera mode. Zoom by pressing and holding the right mouse button (RMB) or MMB, while moving the mouse.

The RMB brings up a context-sensitive menu, with options for editing your selected (red) elements. *Ctrl+RMB* also activates camera mode (useful for those with 2-button mice). If you are in advanced menus, then clicking the RMB on a context menu item with an asterisk (\*) next to it will give you an advanced mesh editing option. Details are visible at the bottom of the Wings 3D screen.

## Free-form vs. exact modelling

In free-form modelling, you are not restricted to depending only on the background image displayed in the viewport to model the object, but are free to make your own changes. Note that most modellers in the market offer only exact modelling, since they lack the flexibility of free-form modelling. Only one or two commercial free-form modellers have been successful, like LightWave 3D, Zbrush and Mudbox.

Wings 3D can be used for both free-form as well as exact modelling.

## An inspiring modeller

Those who have delved deep into 3D modelling, and particularly organic modelling, will find that creating models of creatures, animals and humans is a great challenge to the 3D artist. Not many 3D applications offer a free-form modelling approach like Wings 3D does. The tools in Wings 3D lend themselves to the dexterity of the artist. What is especially sound about Wings 3D is that all the elements are flexible. Extrusion is possible not just at the face level, but also at the edge and the vertex level. Similarly, beveling is also possible in all three modes. Another great advantage of Wings 3D as an organic

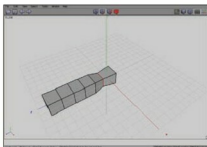


Figure 12: Penultimate loop scaled down.

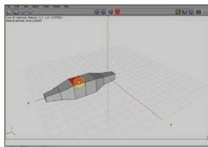


Figure 15: Face mode, third top face selected

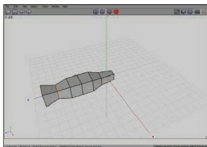


Figure 13: Second edge loop scaled down.

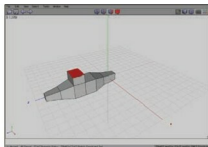


Figure 16: Top face extruded once

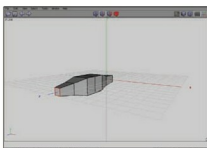


Figure 14: First edge loop scaled down.

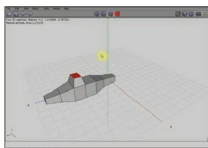


Figure 17: Top face scaled down